Safety Science 117 (2019) 447–457

Contents lists available at ScienceDirect

Safety Science

journal homepage: www.elsevier.com/locate/safety

Walking behavior of pedestrian social groups on stairs: A field study

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ARTICLE INFO

Keywords:
Pedestrian groups
Walking behavior
Movement dynamics
Stairs
Field study

ABSTRACT

There is growing consensus that most pedestrians in a crowd walk in groups, and these groups have an influence on crowd dynamics whenever under natural or emergency situations. Despite this fact, however, empirical data on group walking behavior on stairs are limited. This paper aims to explore how group members interact with each other and how groups organize on stairs. In order to achieve these, a field study was conducted on stairs in a campus of a university. A total of 105 pedestrian groups descending the stairs were selected as observation subjects. After obtaining their trajectories by the optimal flow algorithm, interpersonal distances and angles between group members, average speed, space-time diagrams offset angles and walking dissimilarity on stairs were calculated and analyzed. Results demonstrate that average distances for groups of different sizes are relatively stable. Groups with different sizes on stairs can form different walking patterns. Members belonging to the same group always seek to move at similar speed and offset angles, and can adjust their motion during the descending process, in order to ensure comfortable group walking and coherence. It is discovered that group size has a significant negative effect on average group speed. Gender and social relationships have little influence on group walking speed and dissimilarity values, except in dyads. These findings have implications for group walking modelling and safe pedestrian facility design.

1. Introduction

With the increasing number of mass events or emergency situations in recent years, how to ensure crowd safety has drawn significant interest and attention. Human behavior and evacuation dynamics are becoming important research objectives. In fact, social interactions among pedestrians play a role in the complex dynamics of crowd behavior. It is necessary to investigate them for a deep understanding of crowd dynamics.

Pedestrian social groups (von Krüchten and Schadschneider, 2017) can be always observed in our daily life. Persons may choose to conduct activities (such as shopping, working or travelling) with their family members, friends, colleagues, etc. Here, pedestrians in a small group are those who have social relationships, and walk together (Moussaïd et al., 2010). The walking patterns of groups include walking side-by-side, the 'V' shape in triads, 'U' shape in four-person groups and 'river-like' formation (Moussaïd et al., 2010; Karamouzas and Overmars, 2012). These walking patterns are affected by crowd density (Moussaïd et al., 2010; Zanlungo et al., 2015; Zanlungo et al., 2017; Zanlungo and Kanda, 2015). It was reported that 70% of pedestrians walked in a group consisting of two to four members (Moussaïd et al., 2010). Pedestrians in a group always seek to adjust their speed, movement direction and interpersonal distances in order to maintain the group formation (Zaki and Sayed, 2018). Considering the large number of pedestrian social groups in a crowd, researchers begin to focus on small pedestrian group behavior.

Both models and experiments have been used to investigate pedestrian social groups (Cheng et al., 2014). Models, such as agent-based models (Fang et al., 2016; Bandini et al., 2014; Gorrini et al., 2018), game-theory models (Huang et al., 2015; Wang et al., 2015), cellular based models (Pereira et al., 2017; You et al., 2016) and physical based models (Singh et al., 2009; Zhang et al., 2018), have been modified by involving grouping behavior. In agent-based models, pedestrians are endowed with different characteristics, and their decision making can be affected by personal desires (Fu et al., 2016). However, the models are usually computationally expensive than others. In game-theory models, pedestrians evaluate all available chances, and choose one to ensure that its utility is maximum. Everyone's final utility payoffs will rely on the options selected by all pedestrians (Zheng et al., 2009). The game-theory models have been employed to simulate the exit selection behavior. Cellular based models simulate pedestrian movement in discrete space and time according to transition probabilities and several

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https://doi.org/10.1016/j.ssci.2019.04.048
Received 18 October 2018; Received in revised form 18 April 2019; Accepted 30 April 2019
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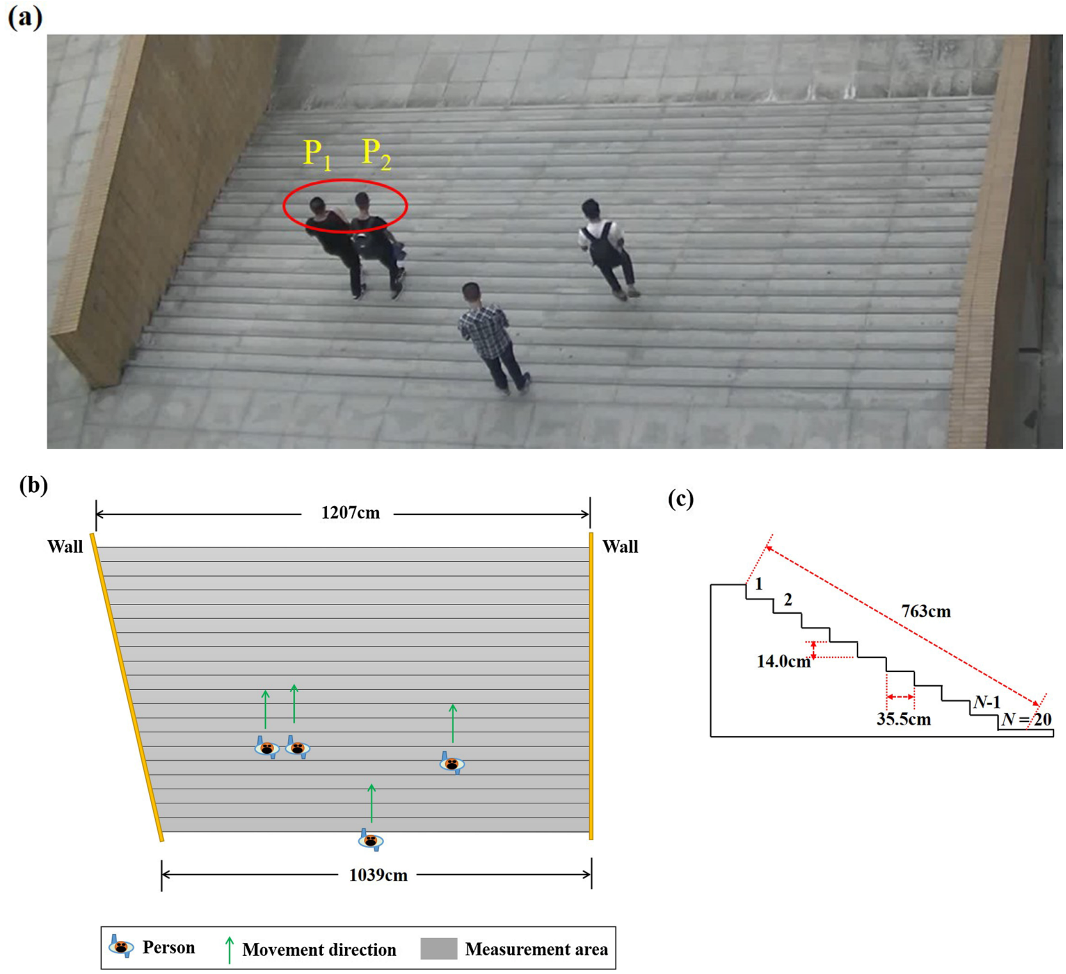


Fig. 1. Experiment setup: (a) A snapshot of the observation area from a video camera; (b) A schematic view of the test area; (c) Dimensions of stairs.

simple rules. They are the most widely adopted to study pedestrian dynamics, especially in complex scenarios (Fu et al., 2016). Physical based models involve optimal acceleration and physical forces among pedestrians, such as the magnetic model (Okazaki and Matsushita, 1993), social force model (Helbing et al., 2000) and centrifugal model (Yu et al., 2005). These models are more suitable to simulate simple scenarios. With all these types of models above, several interesting phenomena and results have been generated. For example, it was found that pedestrian transverse movement during evacuation processes could be blocked by vertical grouping (Wang et al., 2013). Suitable leaders and stable group formation were helpful in reducing collision and evacuation time in a multi-room office with obstacles (Zhang et al., 2018). With the leader-follower evacuation method, more ordering and efficiency displayed during evacuation processes (Dong et al., 2014). It was also reported that the V-like walking pattern of groups was beneficial to social communication among group members. However, this pattern could reduce the flow (Moussaïd et al., 2010).

Experiments are helpful in collecting real movement data of pedestrian groups and providing input for model validation. Many controlled experiments (Okada et al., 2012; Chen et al., 2017; Głapa et al., 2015; Fu et al., 2019) and field studies (Zanlungo et al., 2015; Jazwinski and Walcheski, 2011; Wei et al., 2015) have been conducted to explore group walking behavior on escalators, the effect of pedestrian load or emergency signage on group evacuation, the influences of group sizes, children and density on group locomotion in flat public places, etc. Controlled experiments were primarily performed on stairs (Ma et al., 2017), or in circular arenas (Dyer et al., 2009), corridors (Crociani et al., 2017), rooms (von Krüchten and Schadschneider, 2017) and underground bus concourses (Cuesta et al., 2016). Group behavior primarily during evacuation processes or in bi-directional flow was investigated. It was highlighted that group behavior had a negative influence on the stairwell evacuation performance under normal visibility conditions, but induced an increase in descent speed under impaired visibility conditions (Ma et al., 2017; Zhang et al., 2018). The presence of groups in counter flow could result in fragmented lanes, and reduce specific flow (Crociani et al., 2017).

In field studies, the influences of cultural differences, group size, the number of accompanying children, gender, group composition, etc. on group behavior and crowd dynamics in different public environments were observed. It was demonstrated that both the group size and accompanying children could independently predict pedestrian walking speed in a shopping mall, although neither of them noticeably impacted on movement trajectories (Jazwinski and Walcheski, 2011). In comparison with individual pedestrians on sidewalks, wide-sidewalks and tourist precincts, larger group sizes induced an evident decrease in walking speed (Rastogi et al., 2010). Male groups walked faster than female groups on sidewalks, shopping streets, the seafront, etc. (Costa, 2010). The presence of groups could reduce the capacity of the infrastructure, such as a bridge (Duives et al., 2014). Observation data showed that when dyads crossed non-signalized intersections, there would be a leader who crossed first and then followed by the other one (Gorrini et al., 2017).

From previous research, we can discover that although group behavior has been investigated in different scenarios by experiments, few focused on the movement of social groups on stairs, especially through the field studies. Most research concerning group behavior on stairs was performed under evacuation situations by controlled experiments. Social interactions between group members and microscopic characteristics of group movement on stairs under normal situations were rarely involved. It is well-known that stairs represent the primary egress component for buildings (Ma et al., 2012; Huo et al., 2016). Pedestrian movement on stairs is more complicated than that in corridors or rooms (Ma et al., 2017). There is a need to systematically investigate how pedestrian group members interact with the others and what the feature of their organization is on stairs. These will be useful for a better understanding of group behavior in different scenarios, and for crowd management.

This paper aims to explore intragroup organization and microscopic movement characteristics (i.e., speed, space-time diagrams and offset angles) on stairs, and to compare the difference in social interactions among group members on stairs with that in/on passageways, sidewalks, etc. The influences of pedestrian characteristics (e.g., gender and social relationships) on group movement are further discussed. Theseare performed through a field study on stairs outside a building located in a university. The remainder of this paper is organized in the following: Section 2 presents the detailed experiment setup; Section 3 introduces experimental results and discussions; Finally, conclusions are reported in Section 4.

2. Experiment setup

2.1. Experiment description

The video-recorded observation was conducted on stairs outside a 5-storey teaching building in a campus of a Chinese university on a weekday (9:30 a.m. – 12:30p.m.). The stairs connected the ground with the second floor. Fig. 1 depicts the scenario and dimensions of the stairs. In our experiment, the observation area only comprised a flight of stairs with a total of 20 steps near the second floor. The stairs are restricted by walls on two sides (see Fig. 1(a)), forming a trapezoid if viewing from the top of the stairs (see Fig. 1(b)). The upper and bottom widths of the trapezoid are respectively 1207 and 1039 cm. The tread of each stair is 35.5 cm deep, and the riser of each stair is 14.0 cm high (see Fig. 1(c)).

As the observation area was connected with the teaching building, subjects of the field study were groups of healthy college students, whose ages were approximately between 18 and 26 years old, and heights were approximately between 150 and 180 cm. Here, we focus on the descending movement process of different groups on stairs (see Fig. 1(b)). The density was not very large, and thus group members' physical contact with out-group pedestrians could be avoided. The identification of social groups was according to pedestrians' locomotion, verbal and non-verbal behaviors, as used in Refs. (Zanlungo et al., 2015; Costa, 2010; Gorrini et al., 2017; Zanlungo et al., 2014). More specifically, locomotion behavior primarily included high spatial cohesion and coordination when moving, waiting dynamics to regroup to avoid separation, and leader/follower dynamics during direction changes. Verbal behavior included talking during walking. Non-verbal behavior consisted of physical contact, body and gaze orientation to each other and gesticulation when talking or indicating. Consequently, a total of 105 groups were selected as the observational sample, namely 33 (31.4%) male dyads, 33 (31.4%) female dyads, 5 (4.8%) couples (i.e., "lovers"), 14 (13.3%) male triads, 14 (13.3%) female triads, 3 (2.9%) four-male groups and 3 (2.9%) four-female groups (see Table 1). In order to conveniently distinguish a member from the others in the same group, group members were numbered P_1, P_2, … , P_k from left to right of video images, as shown in Fig. 1(a). Here, k is the total number of members in a group.

2.2. Data collection

The descending movement processes of all groups were recorded by a video camera with a resolution of 1920 × 1080 pixels and a frame rate of 25 fps. The camera was placed on the fifth floor of the teaching building. In order to analyze movement characteristics among group members, it is necessary to first detect and track pedestrians in each frame of the video recordings.

Table 1
The number of different groups.

Group type	The number (percentage)
Male dyads	33 (31.4%)
Female dyads	33 (31.4%)
Couples	5 (4.8%)
Male triads	14 (13.3%)
Female triads	14 (13.3%)
Four-male groups	3 (2.9%)
Four-female groups	3 (2.9%)

Three procedures were involved to collect movement data (i.e., X and Y coordinates at each frame) as follows. The first procedure was the identification of social groups from video images according to that described in Section 2.1, as many individuals also walked on the stairs. The second procedure was to automatically extract each group member's trajectory from the video recordings with the optimal flow method (Wei et al., 2015; Barron et al., 1994), which is a useful image processing technique for detecting moving objects in a scenario where its detailed features are not needed to be acquired in advance. In this way, everyone's trajectory in image space could be gained. Because of the perspective projection and camera lens distortion (see Fig. 1(a)), in the third procedure, trajectories of group members in image space should be corrected and transformed to those in real space. For this purpose, a direct linear transformation method (Ma et al., 2010; Chen et al., 2017) was adopted. Here, real space was established in an elevated perspective plane which was parallel to the incline direction of the stairs (see the blue dashed outline in Fig. 2). According to three phases described above, an example of group members' trajectories in real space is illustrated in Fig. 3.

After obtaining group members' coordinates in real space at each frame, other parameters such as interpersonal distances and angles between two neighboring group members, walking speed, space-time relationship and offset angles on stairs could be calculated. Detailed analysis will be presented in the next section.

3. Results and discussion

3.1. Interpersonal distances and angles in group walking on stairs

Interpersonal distances and angles in group walking (see Fig. 4) (Moussaïd et al., 2010) are important variables, which can be employed to explore the patterns of spatial organization of pedestrian social groups under different conditions. Previous observations have revealed interpersonal distances and angles between group members such as in passageways or other public places (Moussaïd et al., 2010; Wei et al., 2014). However, interpersonal distances, angles and group structure on stairs, especially under normal situations, have not been well investigated. Therefore, we calculate interpersonal distances (d_{ij}) and angles (α_{ij}) between two neighboring members (P_i and P_j) in pedestrian groups of different sizes on stairs, as displayed in Tables 2 and 4. The differences of interpersonal distance between categories with different group sizes are provided. The distributions of interpersonal angles for different group sizes are also analyzed. Here, group member P_j is the closest neighbor of member P_i on his/her right-hand side, and they are in the same group (see Fig. 4). The direction of angles at each frame is calculated according to pedestrians' locations on stairs and the y-axis orthogonal to steps.

From Table 2, it is evident that average distances (d_{ij}) for different group sizes on stairs in this paper range between 0.75 and 0.87 m, i.e.

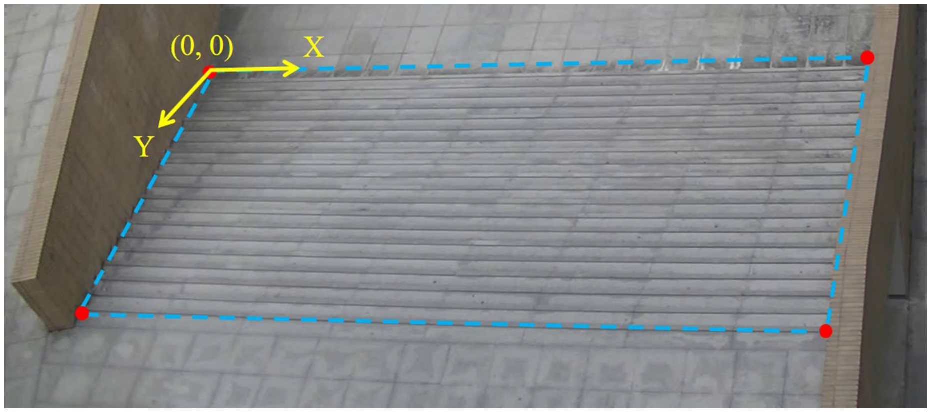


Fig. 2. The incline direction of the stairs represented by the blue dashed outline. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

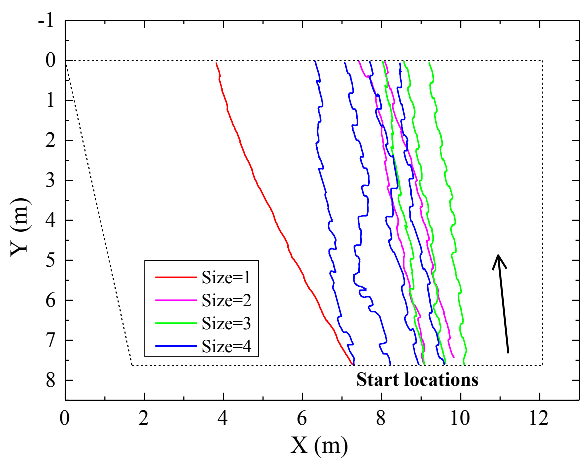


Fig. 3. An example of four groups' trajectories with different group sizes in real space. The coordinate directions correspond to those in Fig. 2. Start locations represent starting points during group members' descending movement processes.

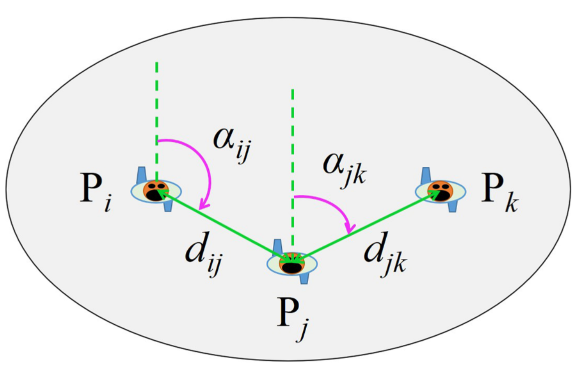


Fig. 4. The distance (d_{ij} or d_{jk}) and angle (α_{ij} or α_{jk}) between pedestrians P_i and P_j or P_j and P_k in a group. Here, P_j (P_k) is the closest neighbor of P_i (P_j) on his/her right-hand side.

the average distances are relatively stable in order to maintain group structure. In comparison with data gained through observations in a passageway of a campus and a public place of a city at low density respectively by Wei et al. (2014) and Moussaïd et al. (2010), the values of d_{ij} in this paper are a little larger than those presented by Wei et al., while they are close to results introduced by Moussaïd et al. It was highlighted that crowd density, body size, culture and social relationship may result in the difference in the values of d_{ij} (Wei et al., 2014).

The comparison of interpersonal distance in groups of different sizes is shown Table 3. Here, the statistical test, namely Analysis of Variance (ANOVA) (Zanlungo et al., 2017), is adopted. The interpersonal distances between P_2 andP_3 in triads or P_3 and P_4 in four-person groups are all compared with those in dyads. From Table 3, it is observed that the difference of interpersonal distances in groups of different sizes is not statistically significant, as reflected by corresponding low ANOVA *F* and high *p* values ($p > 0.05$).

Table 4 shows the average angles (α_{ij}) between neighboring group members for each group size during the descending movement on stairs. Their distributions for different group sizes are depicted in Fig. 5. It was observed that most pedestrians in dyads tended to walk abreast or in ladder shapes ($\alpha_{ij} = 80.22°$). This can be also revealed from the distribution of α_{12} in Fig. 5(a). The average interpersonal angle in dyads on stairs is smaller than that on flat ground reported by Wei et al. or Moussaïd et al.

In groups of size 3, a 'V' shape of walking patterns frequently appeared, i.e. α_{12} and α_{23} are respectively larger and smaller than 90°, as also discovered on flat ground (Zanlungo et al., 2015; Costa, 2010; Zanlungo et al., 2014; Schultz et al., 2014). However, versa 'V' and ladder shapes also existed during group walking on stairs, as shown by the distributions of α_{12} and α_{23} respectively in Fig. 5(b) and (c). The average angle for triads is smaller than that presented by Wei et al. or Moussaïd et al. It is noteworthy that Wei et al. and Moussaïd et al. gained their results from flat public places. Nevertheless, our observation was conducted on stairs in a campus where subjects were college students. Differences in the social status, culture and facilities may be primary reasons, as emphasized by Wei et al. In order to discover whether the three-person group structure (e.g., 'V' shape or versa 'V' shape) on stairs is symmetric or not (Zanlungo et al., 2015), we also compare interpersonal angles ($180° - \alpha_{12}$) and α_{23} in triads. The distribution of the comparison results satisfying ($180° - \alpha_{12}$) > α_{23} or ($180° - \alpha_{12}$) < α_{23} is displayed in Table 5. It is apparent that the frequency is larger (0.75) when ($180° - \alpha_{12}$) > α_{23}, i.e. the structure in triads is left–right asymmetric. This asymmetry is stronger than that discovered on flat ground by Zanlungo et al. (2015).

In groups of size 4, besides a 'U'-like formation in some groups, most pedestrian groups on stairs tended to be subdivided into dyads (see Fig. 5(d)–(f)), i.e. two group members walked in the front, while the other two walked in the back of the group for the convenience of better communication and collision avoidance. This phenomenon of group splitting could be also observed on flat ground (Moussaïd et al., 2010; Zanlungo et al., 2015). This also demonstrates that groups can form a flexible structure under different conditions.

From the group structure on stairs compared with that on flat ground, it is noteworthy that the ladder shapes in dyads or triads and left-right asymmetry in triads may be due to the effect of stairs.

Table 2
Average distances between group members for each group size.

Group size	Neighbors	d_{ij} (m)		
		In this paper	(Wei et al., 2014)	(Moussaïd et al., 2010)
2	P_1P_2	0.78 (± 0.03)	0.68 (± 0.16)	0.78 (± 0.02)
3	P_1P_2	0.77 (± 0.04)	0.57 (± 0.10)	0.79 (± 0.05)
	P_2P_3	0.80 (± 0.04)	0.56 (± 0.19)	0.81 (± 0.10)
4	P_1P_2	0.83 (± 0.08)	0.61 (± 0.10)	0.87 (± 0.06)
	P_2P_3	0.87 (± 0.11)	0.58 (± 0.10)	0.93 (± 0.09)
	P_3P_4	0.75 (± 0.05)	0.57 (± 0.08)	0.80 (± 0.05)

Note: Values between brackets represent the standard error of the mean.

Table 3
Interpersonal distance dependence on group size.

Group size	Interpersonal distance d_{ij} (m)		
	P_1P_2	P_2P_3	P_3P_4
2	0.78 (± 0.03)		
3	0.77 (± 0.04)	0.80 (± 0.04)	–
4	0.83 (± 0.08)	0.87 (± 0.11)	0.75 (± 0.05)
F	0.13	0.39	0.05
p	0.88	0.68	0.82

Note: Values between brackets represent the standard error of the mean. F and p respectively represent ANOVA F and p values.

However, it is not clear if they are affected by the asymmetrical studied area or constraint of stairs. This will be further explored in future work.

3.2. Average speed and space-time diagrams

In order to analyze characteristics of group movement on stairs, average descent speed of each group member and space-time diagrams for each group size are derived. Here, the average descent velocity of each pedestrian is the Y component in Fig. 2, i.e. it is parallel to the incline direction of stairs, and its value can be obtained through Eqs. (1) and (2) (Wei et al., 2014) as follows.

$$V(i,t) = \frac{1}{\Delta t}\sqrt{(x(i,t+\Delta t/2)-x(i,t-\Delta t/2))^2+(y(i,t+\Delta t/2)-y(i,t-\Delta t/2))^2} \quad (1)$$

$$\bar{V}_i = \frac{1}{T_i}\sum_{t=0}^{T_i} V(i,t) \quad (2)$$

where $V(i, t)$ is pedestrian i's instantaneous speed at time t. $x(i, t)$ and $y(i, t)$ are coordinates of pedestrian i at time t on stairs. Δt denotes the time interval which equals 0.16 s (i.e., the interval of five frames). T_i is the total time spent by pedestrian i during the descending process. $\bar{V}_i$ is the average speed of pedestrian i on stairs.

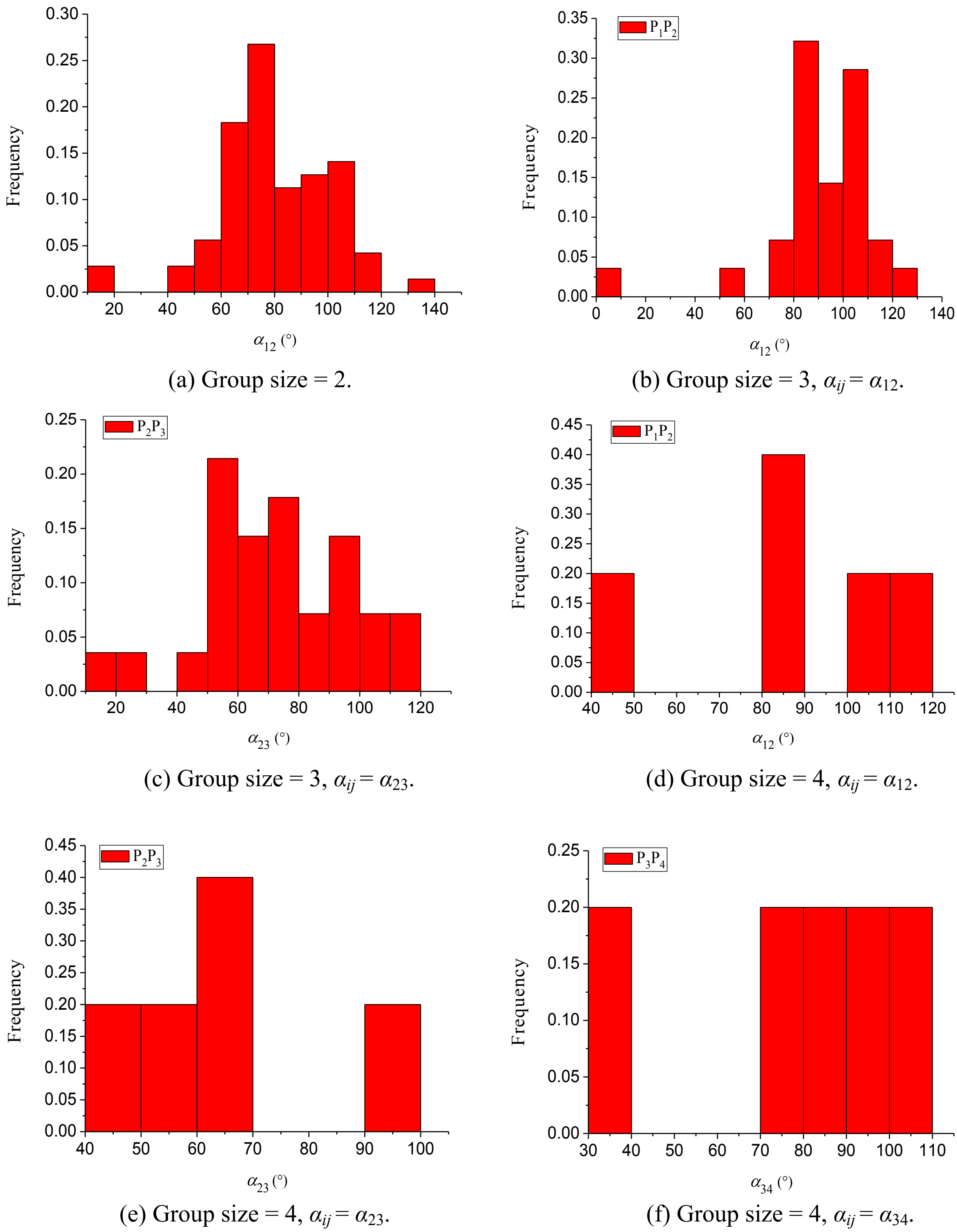
Fig. 6 shows average descent speed of each group member ($\bar{V}_i$) in dyads, triads and four-person groups. A serial number is given to each group. It is apparent that average speeds of pedestrians in different groups fluctuate at the mean of 0.87 (± 0.18) m/s, which is smaller than that in passageways (Wei et al., 2015). More specifically, the mean values of $\bar{V}_i$ in dyads, triads and four-person groups are respectively 0.89 (± 0.20), 0.86 (± 0.14) and 0.77 (± 0.12) m/s in Table 6. The difference is statistically significant, as reflected from high ANOVA F and low p values ($p = 0.01 < 0.05$). The difference can be also found in distributions of speed for different group sizes in Fig. 7. This proves that larger group sizes can induce a decrease in group members' speed on stairs, as also discovered on sidewalks, precincts and so on (Rastogi et al., 2010). Nevertheless, for the same group, there is little difference in group members' average speed between each other. According to calculation, the mean difference values are 0.05 (± 0.06), 0.08 (± 0.05) and 0.15 (± 0.06) m/s in dyads, triads and four-person groups, respectively. Their average values and standard deviations are smaller than the overall standard deviations (± 0.20). It can be deduced that group members always seek to maintain similar speed in order to create comfortable group walking and coherence.

Fig. 8 depicts space-time diagrams for groups of sizes 2–4 during the descending movement process on stairs. It can be observed that group members initially tended to walk side-by-side on the same step. After a period of time, due to a change in patterns of group organization and the stair step constraint, they may walk in different steps, and then maintain a walking pattern with unchanged step intervals. This phenomenon is clear from Table 7 which shows Y (i.e., y-coordinate in real space in Fig. 2, and $Y > 4$ or $Y < 4$) dependence of interpersonal distances in y-coordinate for a dyad ($Y_{g2} = |y_1 - y_2|$), triad ($Y_{g3} = (|y_1 - y_2| + |y_2 - y_3|)/2$) and four-person group ($Y_{g4} = (|y_1 - y_2| + |y_2 - y_3| + |y_3 - y_4|)/3$). Here, as the total length of steps in y-coordinate in real space in Fig. 2 is around 8 m (see Fig. 1(c)), we select $Y > 4$ and $Y < 4$ to represent two parts of the stairs. It is evident that the values of Y_{g2}, Y_{g3} and Y_{g4} in the first several steps (i.e. $Y > 4$) are smaller than those in the latter several steps (i.e. $Y < 4$), as revealed by corresponding high ANOVA F and low p values ($p < 0.05$) in Table 7. Moreover, this difference is more significant in the four-person group. These results also suggest that intragroup members have the capability to adjust their movement after a period of time in order to recreate a stable group structure on stairs.

Table 4
Average angles between group members for each group size.

Group size	Neighbors	α_{ij} (°)		
		In this paper	(Wei et al., 2014)	(Moussaïd et al., 2010)
2	P_1P_2	80.22 (± 2.51)	89.20 (± 15.36)	89.80 (± 1.12)
3	P_1P_2	90.61 (± 4.26)	102.07 (± 20.05)	97.80 (± 5.14)
	P_2P_3	73.29 (± 4.81)	69.00 (± 20.05)	87.10 (± 4.46)
4	P_1P_2	88.72 (± 11.71)	110.41 (± 15.17)	99.20 (± 6.33)
	P_2P_3	65.65 (± 8.52)	91.99 (± 30.03)	87.70 (± 6.54)
	P_3P_4	80.95 (± 12.42)	73.21 (± 23.61)	85.40 (± 5.01)

Note: Values between brackets represent the standard error of the mean.



(a) Group size = 2.
(b) Group size = 3, $\alpha_{ij} = \alpha_{12}$.
(c) Group size = 3, $\alpha_{ij} = \alpha_{23}$.
(d) Group size = 4, $\alpha_{ij} = \alpha_{12}$.
(e) Group size = 4, $\alpha_{ij} = \alpha_{23}$.
(f) Group size = 4, $\alpha_{ij} = \alpha_{34}$.

Fig. 5. Distributions of interpersonal angles (α_{ij}) for different group sizes.

Table 5
The distribution of interpersonal angles satisfying (180° − α_{12}) > α_{23} or (180° − α_{12}) < α_{23} in triads.

Conditions	Frequency
(180° − α_{12}) > α_{23}	0.75
(180° − α_{12}) < α_{23}	0.25

3.3. Offset angles on stairs

Besides average speed and space-time diagrams described above, it is also necessary to explore the consistency of group walking from the perspective of their movement direction on stairs. Here, offset angles (θ_1 or θ_2 in Fig. 9) are employed, which reflect offset in the lateral direction from pedestrian' start location (when entering the measurement area) to an end point (when leaving the measurement area) on stairs. They can be gained through Eqs. (3) and (4) in the following.

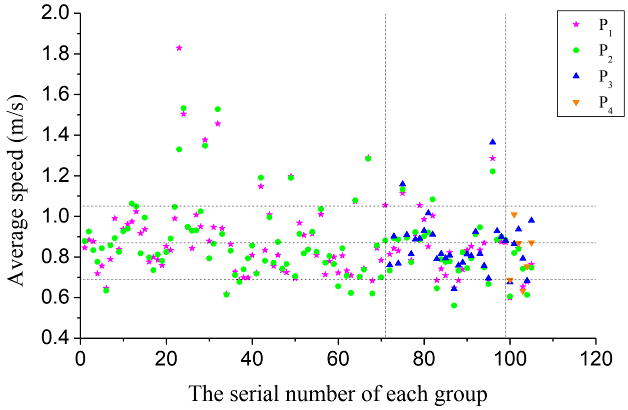


Fig. 6. Average speed of each member (P_k, k = 1, 2, 3, 4) in groups of sizes 2–4 during the descending process. Serial numbers 1–71, 72–99 and 100–105 respectively represent dyads, triads and four-person groups, and they are separated by two vertical dashed lines. The mean and standard deviation of all average speed values (namely 0.87 (± 0.18) m/s) are marked by three horizontal dashed lines.

Table 6
Average speed dependence on group size.

Group size	$\bar{V}_i$ (m/s)
2	0.89 (± 0.20)
3	0.86 (± 0.14)
4	0.77 (± 0.12)
F	4.40
p	0.01

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

$$\theta_1 = \arctan(\Delta X_1/\Delta Y) \quad (3)$$

$$\theta_2 = \arctan(\Delta X_2/\Delta Y) \quad (4)$$

where ΔX_1 (or ΔX_2) is the difference between the *x*-values of the start location and end point 1 (or end point 2) in real space in Fig. 2. ΔY denotes the difference between the *y*-values of the start location and end points 1 or 2. Here, $\theta_1 \geq 0°$, and $\theta_2 \leq 0°$.

Fig. 10 illustrates the offset angle of each member in all groups on stairs. It is noticeable that most pedestrians' offset angle ranges between 0° and 20°, as most pedestrians' destination is an entry to a road of the campus near the left side of the stairs. Moreover, from Table 8, it is observed that the difference of offset angles in groups of different sizes is not statistically significant, as reflected by corresponding low ANOVA *F* and high *p* values (p = 0.13 > 0.05). For the same group, there is slight difference in the offset angles of group members. The average difference values are 1.52 (± 1.59), 4.54 (± 3.96) and 3.33 (± 1.80) degrees in dyads, triads and four-person groups, respectively. These values are small in comparison with corresponding average offset angles in Table 8. This demonstrates that group members always follow the other members belonging to the same group, in order to be not separate from the group on stairs.

According to the analyses above, it is evident that there is a good coherence among intragroup members on stairs, from perspectives of interpersonal distances, angles, movement speed and direction. Although group members walk under the effect of stair step constraint, they can maintain a flexible group structure.

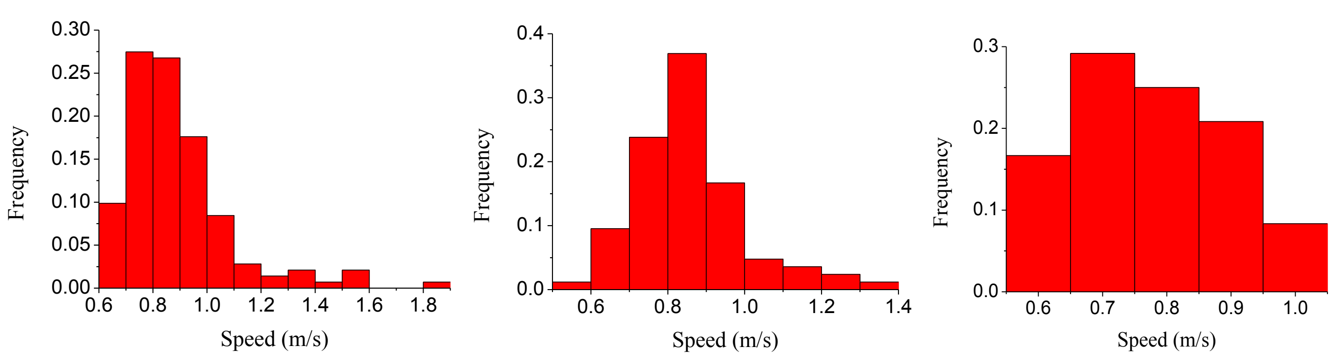
3.4. Influences of pedestrian characteristics

Results above primarily focus on local behavior and movement dynamics of groups on stairs. In this section, we aim to further discuss influences of pedestrian characteristics such as gender and social relationships (friends or couples) on group walking dynamics, along with a comparison with individual walking on stairs. For this purpose, groups and individuals are divided into several categories, and are represented by designations A–I, as listed in Table 9.

The comparison of average distances between group members for different group types during the descending process is illustrated in Fig. 11, Tables 10 and 11. Average distances of groups C–I are 0.82 (± 0.27), 0.74 (± 0.23), 0.75 (± 0.22), 0.79 (± 0.17), 0.78 (± 0.24), 0.84 (± 0.15) and 0.78 (± 0.22) m, respectively. It is apparent that differences in average distance for different group types induced by the gender or social relationships are not statistically significant, as shown by corresponding low ANOVA *F* and high *p* values (p > 0.05) in Tables 10 and 11.

Fig. 12 and Table 12 show average descent speeds for different individual/group types on stairs. Average speeds of A–I are respectively 0.94 (± 0.20), 0.87 (± 0.25), 0.95 (± 0.23), 0.82 (± 0.14), 0.90 (± 0.24), 0.88 (± 0.12), 0.84 (± 0.16), 0.80 (± 0.13) and 0.74 (± 0.10) m/s. It is observed that all-male groups and all-female groups walk at different speeds on stairs, but the difference in walking speed induced by the gender is not statistically significant between these group types, except in dyads (see Table 12). In dyads, couples' average speed is larger than that of two-female groups, but smaller than that of two-male groups. This phenomenon is not consistent with that on flat ground in Ref. (Zanlungo et al., 2017), which showed that the speed distribution for mixed dyads was very similar to that for two-female groups. In comparison with individual walking, group walking will reduce speed, especially in groups of larger sizes. Considering the effect of social relationships in dyads (see Table 13), it is observed that the difference in walking speed between friends and couples is not statistically significant (p > 0.05).

The walking dissimilarity between group members for different



(a) Group size = 2. (b) Group size = 3. (c) Group size = 4.

Fig. 7. The distribution of speed for different group sizes.

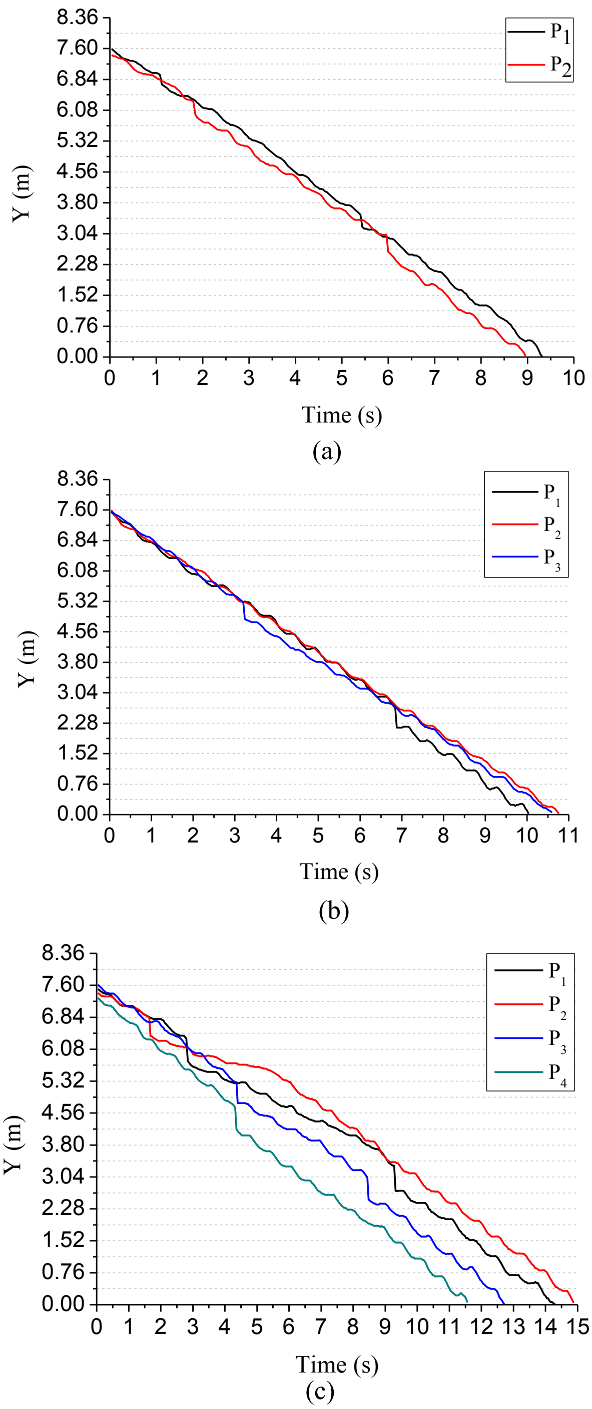


Fig. 8. Space-time diagrams for groups of different sizes: (a) size = 2; (b) size = 3; (c) size = 4. Here, Y is the value of each pedestrian's y-coordinate in real space in Fig. 2. Steps are marked by dashed lines.

Table 7
The comparison of interpersonal distance in y-coordinate in real space (Y_{g2}, Y_{g3} and Y_{g4}) on two parts of the stairs ($Y > 4$ and $Y < 4$).

Y (m)	Y_{g2} (m)	Y_{g3} (m)	Y_{g4} (m)
> 4	0.22 (± 0.12)	0.10 (± 0.07)	0.34
< 4	0.34 (± 0.15)	0.22 (± 0.10)	0.79
F	44.45	107.54	909.58
p	2.04×10^{-10}	0.00	0.00

Note: Values between brackets represent the standard deviation of the mean. F and p respectively represent ANOVA F and p values.

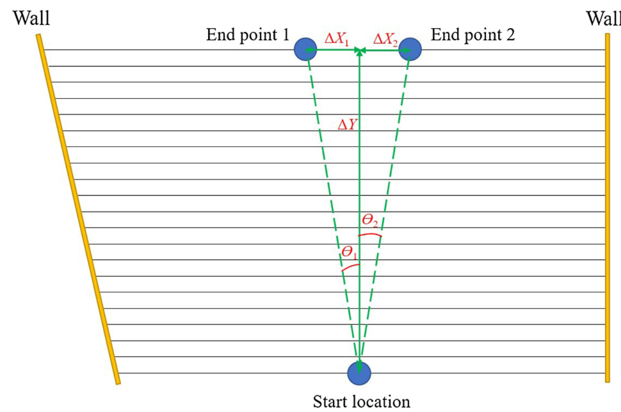


Fig. 9. A schematic illustration of the calculation of offset angles (θ_1 or θ_2) on stairs.

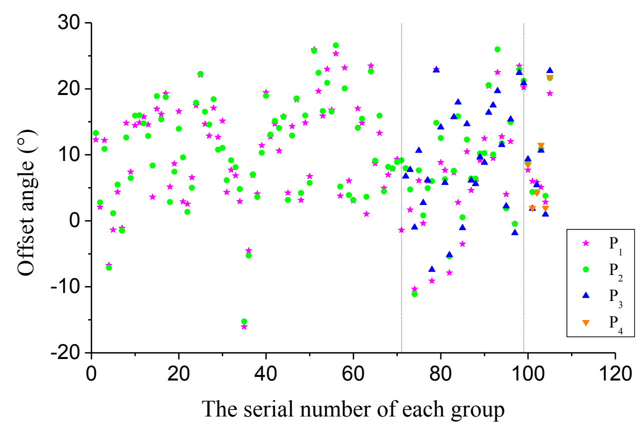


Fig. 10. The offset angle of each member (P_k, k = 1, 2, 3, 4) in different groups during the descending process. Serial numbers 1–71, 72–99 and 100–105 respectively represent dyads, triads and four-person groups, and they are separated by two vertical dashed lines.

Table 8
Offset angle dependence on group size.

Group size	Offset angle (°)
2	10.59 (± 7.82)
3	8.56 (± 8.60)
4	8.38 (± 6.64)
F	2.06
p	0.13

Note: Values between brackets represent the standard deviation of the mean. F and p respectively represent ANOVA F and p values.

Table 9
Designations for different individual/group types.

Designation	Individual/group type
A	Single male
B	Single female
C	Male dyads
D	Female dyads
E	Couples
F	Male triads
G	Female triads
H	Four-male groups
I	Four-female groups

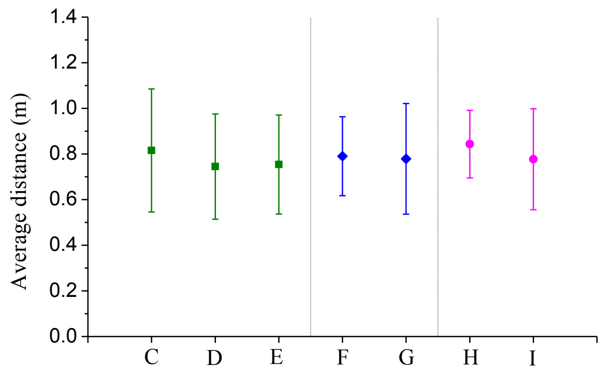


Fig. 11. Average distances between group members for different group types represented by C–I which are listed in Table 9.

Table 10
Average distance dependence on gender for dyads, triads and four-person groups.

Gender	Average distance (m)		
	Dyads	Triads	Four-person groups
All males	0.82 (± 0.27)	0.79 (± 0.17)	0.84 (± 0.15)
All females	0.74 (± 0.23)	0.78 (± 0.24)	0.78 (± 0.22)
Mixed	0.75 (± 0.22)	–	–
F	0.69	0.04	0.49
p	0.50	0.84	0.50

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

Table 11
Average distance dependence on social relationships for dyads.

Social relationships	Average distance (m)
Friends	0.78 (± 0.25)
Couples	0.75 (± 0.22)
F	0.05
p	0.82

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

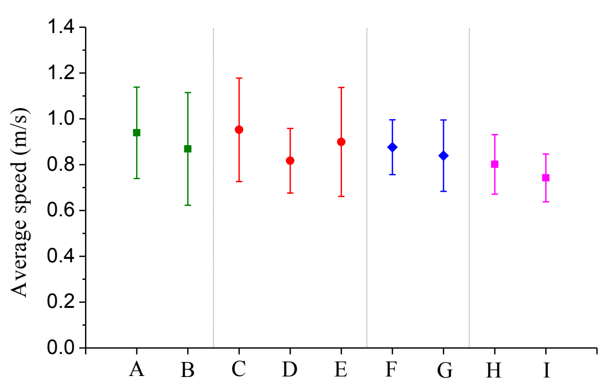


Fig. 12. Average speeds for different individual/group types represented by A–I which are listed in Table 9.

group types on stairs can be also assessed by integrating movement speed with movement direction. It is measured through the following equations (Ma and Song, 2013).

$$d_{ijt} = \alpha\ \|(x, y)_{it} - (x, y)_{jt}\| + \beta\ |v_{it} - v_{jt}| + \gamma\langle (v_x, v_y)_{it}, (v_x, v_y)_{jt}\rangle \tag{5}$$

Table 12
Average speed dependence on gender for singles, dyads, triads and four-person groups.

Gender	Average speed (m/s)			
	Singles	Dyads	Triads	Four-person groups
All males	0.94 (± 0.20)	0.95 (± 0.23)	0.88 (± 0.12)	0.80 (± 0.13)
All females	0.87 (± 0.25)	0.82 (± 0.14)	0.84 (± 0.16)	0.74 (± 0.10)
Mixed	–	0.90 (± 0.24)	–	–
F	1.18	8.16	1.49	1.49
p	0.28	4.45×10^{-4}	0.23	0.23

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

Table 13
Average speed dependence on social relationships for dyads.

Social relationships	Average speed (m/s)
Friends	0.88 (± 0.20)
Couples	0.90 (± 0.24)
F	0.05
p	0.83

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

$$\langle (v_x, v_y)_{it}, (v_x, v_y)_{jt}\rangle = \left(1 - \frac{(v_x, v_y)_{it}(v_x, v_y)_{jt}'}{\sqrt{(v_x, v_y)_{it}'(v_x, v_y)_{it}}\sqrt{(v_x, v_y)_{jt}'(v_x, v_y)_{jt}}}\right) \tag{6}$$

$$\alpha = NTk(k-1)/2 \sum_{n=0}^{N}\sum_{t=0}^{T}\sum_{i=1}^{k-1}\sum_{j=i+1}^{k} \|(x, y)_{it} - (x, y)_{jt}\| \tag{7}$$

$$\beta = NTk(k-1)/2 \sum_{n=0}^{N}\sum_{t=0}^{T}\sum_{i=1}^{k-1}\sum_{j=i+1}^{k} |v_{it} - v_{jt}| \tag{8}$$

$$\gamma = NTk(k-1)/2 \sum_{n=0}^{N}\sum_{t=0}^{T}\sum_{i=1}^{k-1}\sum_{j=i+1}^{k} \langle (v_x, v_y)_{it}, (v_x, v_y)_{jt}\rangle \tag{9}$$

$$d_g = \frac{2}{Tk(k-1)} \sum_{t=0}^{T}\sum_{i=1}^{k-1}\sum_{j=i+1}^{k} d_{ijt} \tag{10}$$

Here, d_{ijt} denotes the walking dissimilarity between group members i and j at time t. d_g in Eq. (10) is the average value of d_{ijt} for a group during the whole descending process. A larger value of d_g represents a higher walking dissimilarity between group members. $(x, y)_{it}$ (or $(x, y)_{jt}$) is the x- and y-coordinates of group member i (or j) in real space (see Fig. 2) at time t. v_{it} (or v_{jt}) is group member i's (or j's) speed at time t. v_x and v_y are respectively x and y components of a pedestrian' velocity at time t. α, β and γ are three parameters used to measure the contribution of each item to the value of d_g. In order to ensure the value of d_{ijt} to be a pure number, all these three parameters are defined as the inverse of corresponding average values between pedestrians on the whole set. They are calculated through Eqs. (7)–(9). T is the total period of time spent by a group in descending the stairs. k is the total number of members in a group. N is the total number of groups. It is evident that differences in locations, speed and movement direction among group members are respectively reflected in the first, second and third items in Eq. (5).

Values of d_g for different group types on stairs are depicted in Fig. 13. The mean values of d_g in groups C–I are 10.44 (± 2.43), 9.98 (± 2.60), 8.17 (± 2.18), 10.74 (± 1.12), 11.16 (± 1.94), 11.21 (± 0.65) and 11.16 (± 0.79), respectively. In dyads and four-person groups, it is evident that the mean of d_g for all-male groups is a little

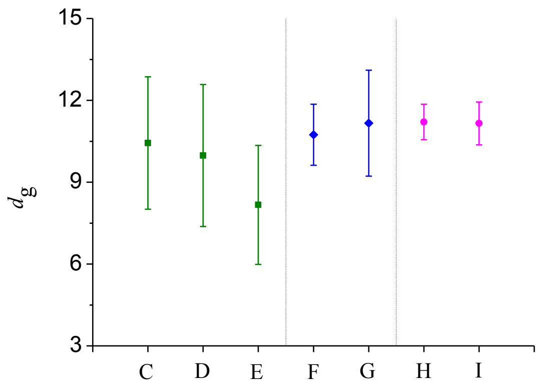


Fig. 13. The walking dissimilarity (d_g) between group members for different group types represented by C–I which are listed in Table 9.

Table 14
Walking dissimilarity dependence on gender for dyads, triads and four-person groups.

Gender	Walking dissimilarity		
	Dyads	Triads	Four-person groups
All males	10.44 (± 2.43)	10.74 (± 1.12)	11.21 (± 0.65)
All females	9.98 (± 2.60)	11.16 (± 1.94)	11.16 (± 0.79)
Mixed	8.17 (± 2.18)	–	–
F	1.83	0.51	0.01
p	0.17	0.48	0.94

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

Table 15
The comparison of walking dissimilarity values between dyads and non-dyads.

	Walking dissimilarity
Dyads	10.06 (± 2.53)
Non-dyads	10.99 (± 1.45)
F	3.95
p	0.049

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

Table 16
Walking dissimilarity dependence on social relationships for dyads.

Social relationships	Walking dissimilarity
Friends	10.21 (± 2.51)
Couples	8.17 (± 2.18)
F	3.12
p	0.08

Note: Values between brackets represent the standard deviation of the mean. *F* and *p* respectively represent ANOVA *F* and *p* values.

larger than that for all-female groups, and the mean of d_g for couples is the smallest. However, the mean of d_g for three-male groups is a little smaller than that for three-female groups. These differences induced by the gender are not statistically significant from our collected data set (see Table 14). By comparing the mean walking dissimilarity values between dyads and non-dyads in Table 15, it is observed that the mean walking dissimilarity value of dyads is smaller than that of non-dyads, and the difference is statistically significant ($p < 0.05$). This suggests that dyads walk on stairs in a more coordinated way. From the perspective of social relationships in dyads, it is observed that the difference in the mean of d_g between friends and couples is not significant (see Table 16).

4. Conclusions

In this paper, an empirical observation was performed on stairs in a campus of a university, in order to investigate group walking behavior and movement dynamics. A video camera was employed to record descending movement processes of groups. Then 105 pedestrian groups were detected and tracked to obtain their trajectories in real space. According to this, interpersonal distances and angles between group members, average descent speed, space-time diagrams, offset angles and the influence of pedestrian characteristics are analyzed and discussed.

Experimental findings reveal that average distances for groups of different sizes are relatively stable on stairs for the purpose of maintaining group structure, although their values may vary in different types of facilities. During the descending process, walking side-by-side in dyads, 'V'-like shapes in triads and 'U'-like formation in four-person groups are frequently observed. Besides these, ladder shapes in dyads, versa 'V' and ladder shapes in triads, and four-person groups subdivided into dyads also exist, i.e. pedestrian groups can create a flexible structure, even under the effect of stair step constraint. It is observed that the walking patterns in triads on stairs are left–right asymmetric.

In viewing of group members' average speeds and space-time diagrams on stairs, it is discovered that pedestrians belonging to the same group always tend to descend at similar speed, and can adjust their movement during descending processes, for comfortable group walking and coherence. The negative effect of group size on average group speed is significant. From the analysis of movement direction on stairs, it is demonstrated that there is slight difference in offset angles between each member in the same group, i.e. group members will follow the others in the same group in order to be not separate from the group. The influence of group size on offset angles is not statistically significant.

We further discuss influences of pedestrian characteristics (e.g., gender and social relationships) on group walking dynamics on stairs. Results suggest that the differences in average interpersonal distance for different group sizes induced by gender or social relationships are not statistically significant. The differences in average walking speed for group sizes 3 and 4 induced by gender are also not statistically significant. However, the difference is significant in dyads, i.e. couples' average speed is larger than that of two-female groups, but smaller than that of two-male groups. The effect of social relationships on walking speed in dyads is not significant. The differences in walking dissimilarity values for different group types induced by gender or social relationships in our collected data set are not very significant, but the average walking dissimilarity value of dyads is significantly smaller than that of non-dyads.

As in any experiment, there are limitations and extension which need to be further investigated in future studies. For example, (1) the collected data are limited, especially the numbers of couples, three-person groups and four-person groups. (2) It will be interesting to study the effects of geometries of stairs and density on group walking patterns and dynamics. (3) We just focus on group members' descending processes on stairs, and it will also be very interesting to compare all results with those during ascending processes. Despite these, we believe that our quantitative empirical findings may provide better insight into group movement dynamics on stairs, and be helpful for improvements of group walking modelling.

Acknowledgements

This research was supported by National Natural Science Foundation of China (grant numbers 71804026, 51803031); Major Project Funding for Social Science Research Base in Social Science Planning of Fujian Province (grant number FJ2018JDZ022);

Educational and Scientific Research Program of the Education Department of Fujian Province, China for Young and Middle-aged Teachers (grant number JT180048); Natural Science Foundation of Fujian Province, China (grant number 2018J05078); and the Foundation for Talents of Fuzhou University, China (grant number XRC-17040).

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